

EDGE INTELLIGENCE

LAB-5

PRADHEESH RAJ R

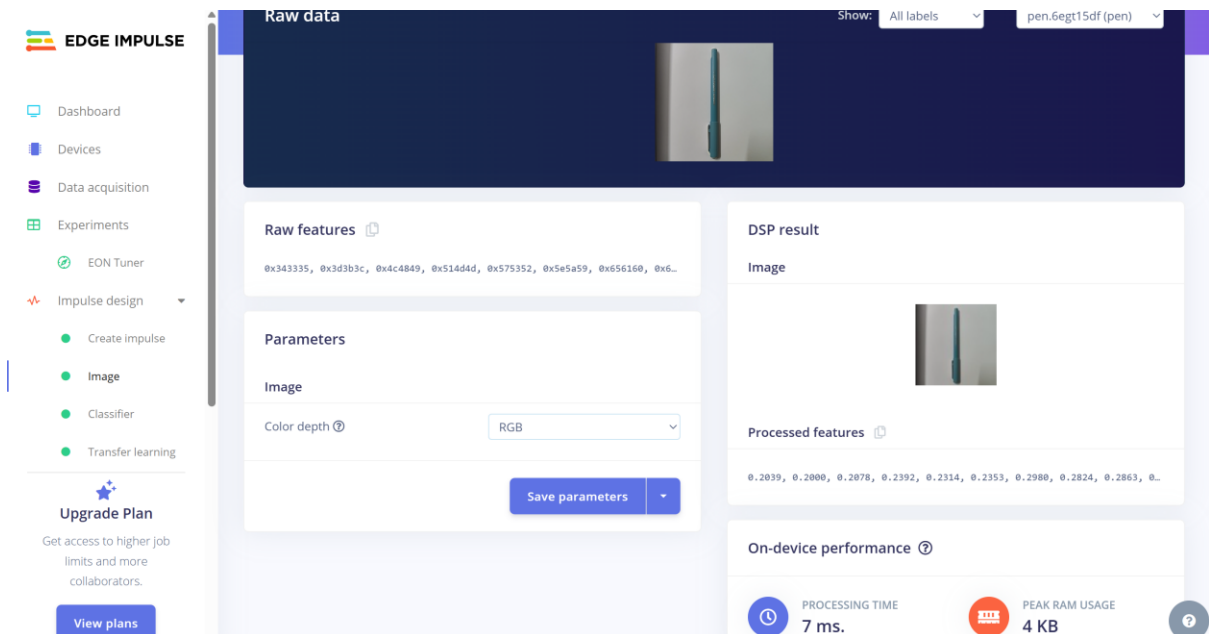
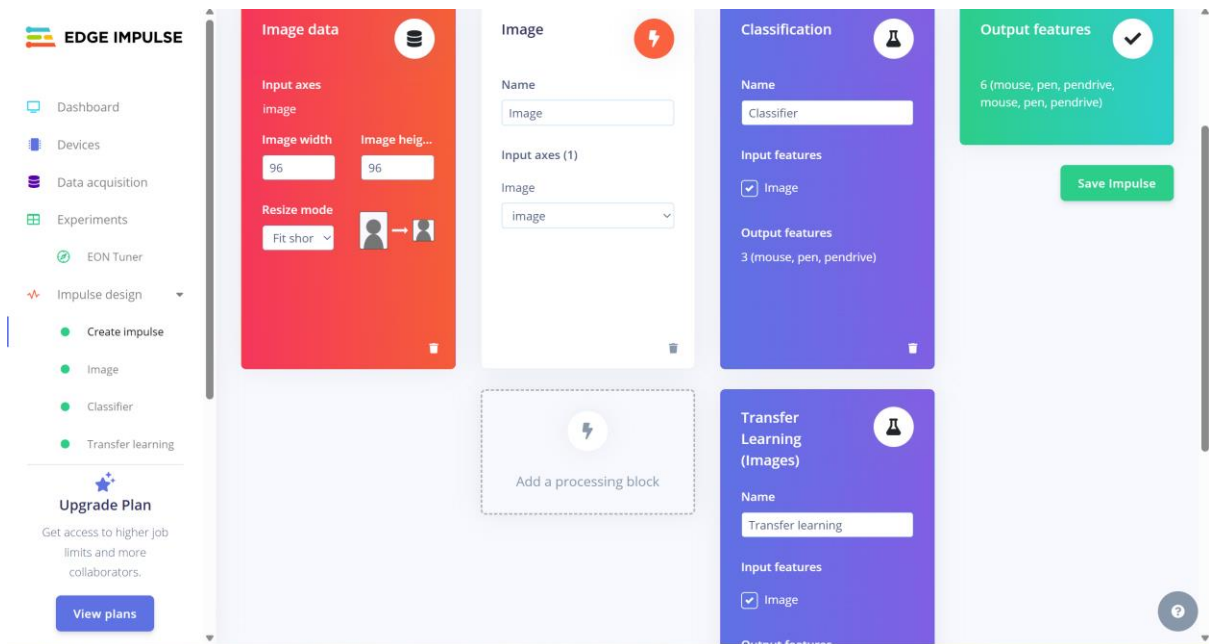
25MML0023

EDGE IMPULSE

- Create account in edge impulse
- Go to data acquisition.
- Choose connect data option.
- Scan QR code using phone.
- Select Label data before clicking a photo.
- Split the clicked photos into train and test data.

The screenshot displays the Edge Impulse web interface. The top navigation bar includes 'Dataset', 'Data explorer', 'Data sources', 'Synthetic data', 'AI labeling', and 'CSV Wizard'. The left sidebar lists various features: Dashboard, Devices, Data acquisition, Experiments, EON Tuner, and Impulse design. The main content area shows a dataset with 15 items collected. A 'TRAIN / TEST SPLIT' section indicates 80% / 20% split. Below this, a table lists dataset items with columns for Sample Name, Label, and Added date. The 'RAW DATA' section shows a photo of a Logitech mouse.

SAMPLE NAME	LABEL	ADDED
mouse.6egt6v78	mouse	Today, 21:02...
mouse.6egt6gnm	mouse	Today, 21:01...
mouse.6egt60mv	mouse	Today, 21:01...
mouse.6egt5fjq	mouse	Today, 21:01...
pendrive.6egt4o0l	pendrive	Today, 21:00...
pendrive.6egt4djn	pendrive	Today, 21:00...
pendrive.6egt3pju	pendrive	Today, 21:00...



EDGE IMPULSE

Dashboard

Devices

Data acquisition

Experiments

EON Tuner

Impulse design

Create impulse

Image

Classifier

Transfer learning

Upgrade Plan

Get access to higher job limits and more collaborators.

View plans

PRADHEESHRAJR / pradheeshrajr_25MMLOO23

PERSONAL

Target: Cortex-M4F 80MHz

Neural Network settings

Training settings

Number of training cycles10

Use learned optimizer

Learning rate0.0005

Training processorCPU

Advanced training settings

Neural network architecture

Neural network

Transfer learning

Input layer (27,648 features)

2D conv / pool layer (16 filters, 3 kernel size, 1 layer)

2D conv / pool layer (32 filters, 3 kernel size, 1 layer)

Training output

Model

Model version: Quantized (int8)

Last training performance (validation set)

ACCURACY33.3%

LOSS0.99

Confusion matrix (validation set)

	MOUSE	PEN	PENDRIVE
MOUSE	100%	0%	0%
PEN	-	-	-
PENDRIVE	0%	100%	0%
F1 SCORE	1.00	0.00	0.00

Metrics (validation set)

METRIC	VALUE
Weighted average Precision	0.33
Weighted average Recall	0.33
Weighted average F1 score	0.33

Neural Network

Transfer learning

Input layer (27,648 features)

2D conv / pool layer (16 filters, 3 kernel size, 1 layer)

2D conv / pool layer (32 filters, 3 kernel size, 1 layer)

Flatten layer

Dropout (rate 0.25)

Add an extra layer

Output layer (3 classes)

Save & train

Metrics (validation set)

METRIC	VALUE
Weighted average Precision	0.33
Weighted average Recall	0.33
Weighted average F1 score	0.33

Data explorer (full training set)

mouse - correct

pen - correct

pendrive - correct

pendrive - incorrect

On-device performance

Engine: EON™ Compiler

INFERENCIN...553 ms.

PEAK RAM U...182.8K

FLASH USAGE88.4K

EDGE IMPULSE

Dashboard

Devices

Data acquisition

Experiments

EON Tuner

Impulse design

Create impulse

Image

Classifier

Transfer learning

Upgrade Plan

Get access to higher job limits and more collaborators.

View plans

PRADHEESHRAJR / pradheeshrajr_25MMLOO23

PERSONAL

Target: Cortex-M4F 80MHz

Neural Network settings

Training settings

Number of training cycles 20

Use learned optimizer

Learning rate 0.0005

Training processor CPU

Data augmentation

Advanced training settings

Neural network architecture

Input layer (27,648 features)

MobileNetV2 96x96 0.35 (final layer: 16 neurons, 0.1 dropout)

Training output

Model

Model version: Quantized (int8)

Last training performance (validation set)

ACCURACY 66.7%

LOSS 0.52

Confusion matrix (validation set)

	MOUSE	PEN	PENDRIVE
MOUSE	100%	0%	0%
PEN	-	-	-
PENDRIVE	50%	0%	50%
F1 SCORE	0.67		0.67

Metrics (validation set)

METRIC	VALUE
Weighted average Precision	0.83
Weighted average Recall	0.67
Weighted average F1 score	0.67

EDGE IMPULSE

Dashboard

Devices

Data acquisition

Experiments

EON Tuner

Impulse design

Create impulse

Image

Classifier

Transfer learning

Upgrade Plan

Get access to higher job limits and more collaborators.

View plans

Neural network architecture

Input layer (27,648 features)

MobileNetV2 96x96 0.35 (final layer: 16 neurons, 0.1 dropout)

Choose a different model

Output layer (3 classes)

Save & train

Metrics (validation set)

METRIC	VALUE
Weighted average Precision	0.83
Weighted average Recall	0.67
Weighted average F1 score	0.67

Data explorer (full training set)

mouse - correct

pen - correct

pendrive - correct

pendrive - incorrect

On-device performance

Engine: EON™ Compiler (RAM optimized)

INFERENCIN... 1008 ms.

PEAK RAM U... 232.9K

FLASH USAGE 546.4K